Question Number : 50 Question Id : 695278275 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

An inductor of 10 mH, capacitor of 0.1 μF and a resistor of 100 Ω are connected in series across an *a.c* power supply 220 V, 70 Hz. The power factor of the given circuit is 0.5. The difference in the inductive reactance and capacitance reactance is $\sqrt{3}\ \alpha\ \Omega$. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	69527823
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527823
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 695278276 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Number of moles and number of molecules in 1.4187 L of SO_2 at STP respectively are

Options :

695278936. $0.1266; 3.812 \times 10^{22}$

695278937. $0.0633; 3.812 \times 10^{22}$

695278938. $0.1266; 7.6238 \times 10^{22}$

695278939. $0.0633; 7.6238 \times 10^{22}$

Question Number : 52 Question Id : 695278277 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

Options :

695278940. 5:1

695278941. 5:0.81

695278942. 5:1.75

695278943. 5:27

Question Number : 53 Question Id : 695278278 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following is correct set of 4 quantum numbers of 19th electron in Chromium (Atomic number = 24) in accordance with Aufbau principle?

Options :

695278944. $n = 3, l = 2, m = +2, s = +\frac{1}{2}$

695278945. $n = 3, l = 2, m = -2, s = +\frac{1}{2}$

695278946. $n = 4, l = 1, m = 0, s = +\frac{1}{2}$

695278947. $n = 4, l = 0, m = 0, s = +\frac{1}{2}$

Question Number : 54 Question Id : 695278279 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure.

Statement II: In a constant volume process, no work is produced and all the heat withdrawn goes into the chaotic motion and is reflected by a temperature increase of the ideal gas.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

695278948. Both Statement I and Statement II are true

695278949. Both Statement I and Statement II are false

695278950. Statement I is true but Statement II is false

695278951. Statement I is false but Statement II is true

Question Number : 55 Question Id : 695278280 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

At T(K), the equilibrium constant of $A_2(g) + B_2(g) \rightleftharpoons C(g)$ is 2.7×10^{-5}. What is the equilibrium constant for $\frac{1}{3}A_2(g) + \frac{1}{3}B_2(g) \rightleftharpoons \frac{1}{3}C(g)$ at the same temperature?

Options :

695278952. $(2.7 \times 10^{-5})^3$

695278953. 6×10^{-2}

695278954. $\sqrt{2.7 \times 10^{-5}}$

695278955. 3×10^{-2}

Question Number : 56 Question Id : 695278281 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In order to oxidise a mixture of 1 mole each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is

Options :

695278956. 3

695278957. 2

695278958. 5

695278959. 7

Question Number : 57 Question Id : 695278282 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the first order reaction R → P.
The fraction of molecules decomposed in the given first order reaction can be expressed as

Options :

695278960. $1 - e^{k_1 t}$

695278961. $1 + e^{k_1 t}$

695278962. $1 + e^{-k_1 t}$

695278963. $1 - e^{-k_1 t}$

Question Number : 58 Question Id : 695278283 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A monoatomic anion (A^-) has 45 neutrons and 36 electrons. Atomic mass, group in the periodic table and physical state at room temperature of the element (A) respectively are

Options :

695278964. 80, 17, liquid

695278965. 81, 16, solid

695278966. 80, 16, gas

695278967. 81, 15, gas

Question Number : 59 Question Id : 695278284 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: The covalency of oxygen is generally two but it can exceed upto four. The oxidation state of oxygen in SO_2 is -2 and in OF_2 it is $+2$.

Statement II: The anomalous behaviour of oxygen when compared to the other elements of group 16 is due to its small size and high electronegativity.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

695278968. Both Statement I and Statement II are true

695278969. Both Statement I and Statement II are false

695278970. Statement I is true but Statement II is false

695278971. Statement I is false but Statement II is true

Question Number : 60 Question Id : 695278285 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The **correct** statements among the following are,

A. Mo(VI) and W(VI) are less stable than Cr(VI).
B. Ce^{4+} and Tb^{4+} are oxidant while Eu^{2+} and Yb^{2+} are reductant.
C. Cm and Am have seven unpaired electrons.
D. Actinoid contraction is greater from element to element than lanthanoid contraction.

Choose the correct answer from the options given below:

Options :

695278972. A and B Only

695278973. C and D Only

695278974. B and D Only

695278975. A and C Only

Question Number : 61 Question Id : 695278286 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Correct statements from the following are

A. Potassium dichromate is an oxidising agent and it oxidises $FeSO_4$ to $Fe_2(SO_4)_3$ in acidic medium.
B. Sodium dichromate can be used as primary standard in volumetric estimation.
C. CrO_4^{2-} and $Cr_2O_7^{2-}$ are interconvertible in aqueous solution by varying the pH of the solution.
D. Cr-O-Cr bond angle in $Cr_2O_7^{2-}$ is 126°.

Choose the correct answer from the options given below:

Options :

695278976. A, B and C Only

695278977. A, C and D Only

695278978. A and C Only

695278979. B and D Only

Question Number : 62 Question Id : 695278287 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I Complex ion		**List-II Calculated spin only magnetic moment (BM)**
A.	$[Cr(H_2O)_6]^{2+}$	I.	3.87
B.	$[Co(H_2O)_6]^{2+}$	II.	5.92
C.	$[Cu(H_2O)_6]^{2+}$	III.	4.90
D.	$[Mn(H_2O)_6]^{2+}$	IV.	1.73

Choose the ***correct*** answer from the options given below:

Options :

695278980. A-I, B-III, C-IV, D-II

695278981. A-II, B-I, C-III, D-IV

695278982. A-IV, B-II, C-I, D-III

695278983. A-III, B-I, C-IV, D-II

Question Number : 63 Question Id : 695278288 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Increasing order of electron withdrawing power of following functional groups is:

a. $-CN$
b. $-COOH$
c. $-NO_2$
d. $-I$

Options :

695278984. c < b < d < a

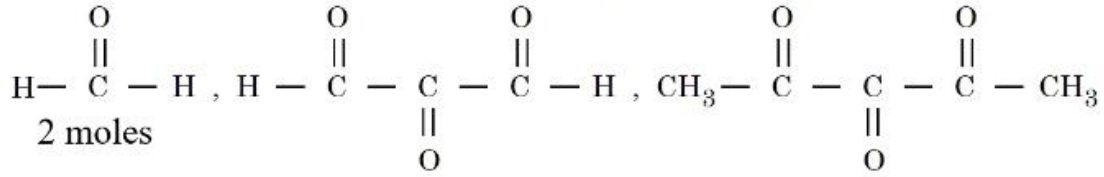
695278985. c < a < b < d

695278986. d < b < a < c

695278987. a < b < c < d

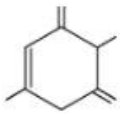
Question Number : 64 Question Id : 695278289 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

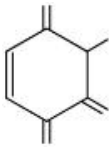
An alkene (X) on ozonolysis followed by reduction gives following products.

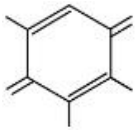


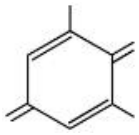
The alkene (X) is:

Options :

695278988. 

695278989. 

695278990. 

695278991. 

Question Number : 65 Question Id : 695278290 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

List-I Name of reaction		List-II Reagent or catalyst used	
A.	Finkelstein reaction	I.	SbF_3
B.	Swarts reaction	II.	Na, dry ether
C.	Sandmeyer's reaction	III.	NaI
D.	Fittig reaction	IV.	Cu_2Cl_2

Choose the ***correct*** answer from the options given below:

Options :

695278992. A-I, B-IV, C-III, D-II

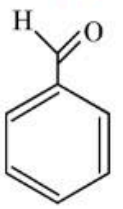
695278993. A-III, B-I, C-IV, D-II

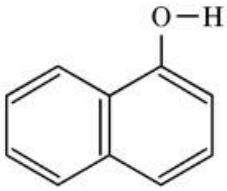
695278994. A-IV, B-II, C-I, D-III

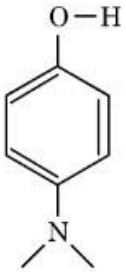
695278995. A-I, B-III, C-II, D-IV

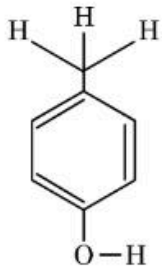
Question Number : 66 Question Id : 695278291 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

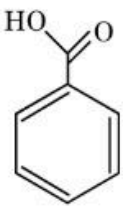
Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is:

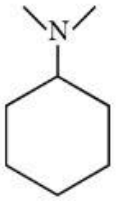

(I)

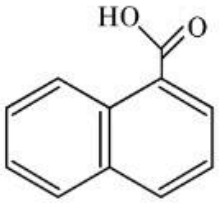

(II)


(III)

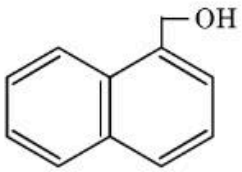

(IV)


(V)


(VI)


(VII)


(VIII)


(IX)

Options :

695278996. 5

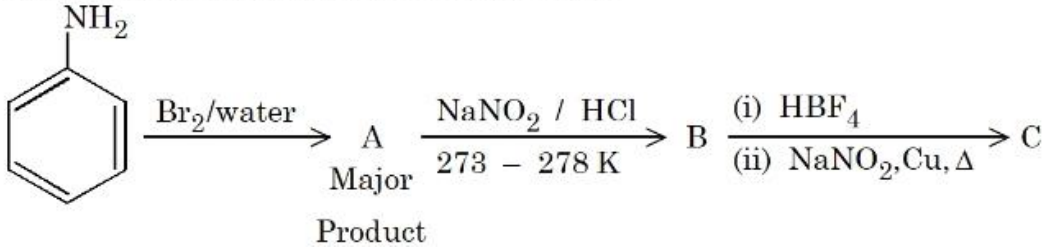
695278997. 4

695278998. 6

695278999. 3

Question Number : 67 Question Id : 695278292 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Product C of the following reaction sequence will be



Options :

6952781000. 1-Bromo-4-nitrobenzene

6952781001. 1, 3, 5-Tribromo-2-nitrobenzene

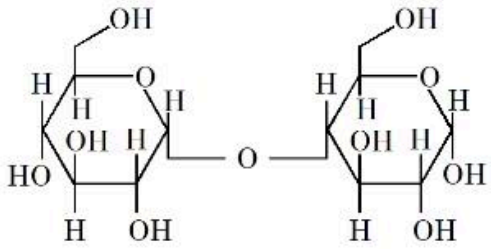
6952781002. 4-Bromo-1-nitrobenzene

6952781003. 1, 3, 5-Tribromobenzene

Question Number : 68 Question Id : 695278293 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

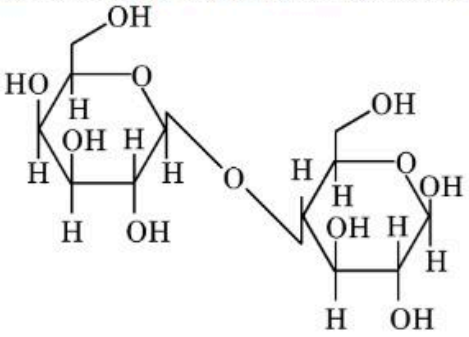
Given below are two statements:

Statement I: The structure of Maltose is given below:



Maltose is a non-reducing sugar.

Statement II: The structure of Lactose is given below:



Lactose is a reducing sugar.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952781004. Both Statement I and Statement II are true

6952781005. Both Statement I and Statement II are false

6952781006. Statement I is true but Statement II is false

6952781007. Statement I is false but Statement II is true

Question Number : 69 Question Id : 695278294 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I Name of amino acid		**List-II One letter symbol/type**
A.	Arginine	I.	D/Non-essential
B.	Aspartic acid	II.	R/Essential
C.	Lysine	III.	E/Non-essential
D.	Glutamic acid	IV.	K/Essential

Choose the ***correct*** answer from the options given below:

Options :

6952781008. A-II, B-I, C-IV, D-III

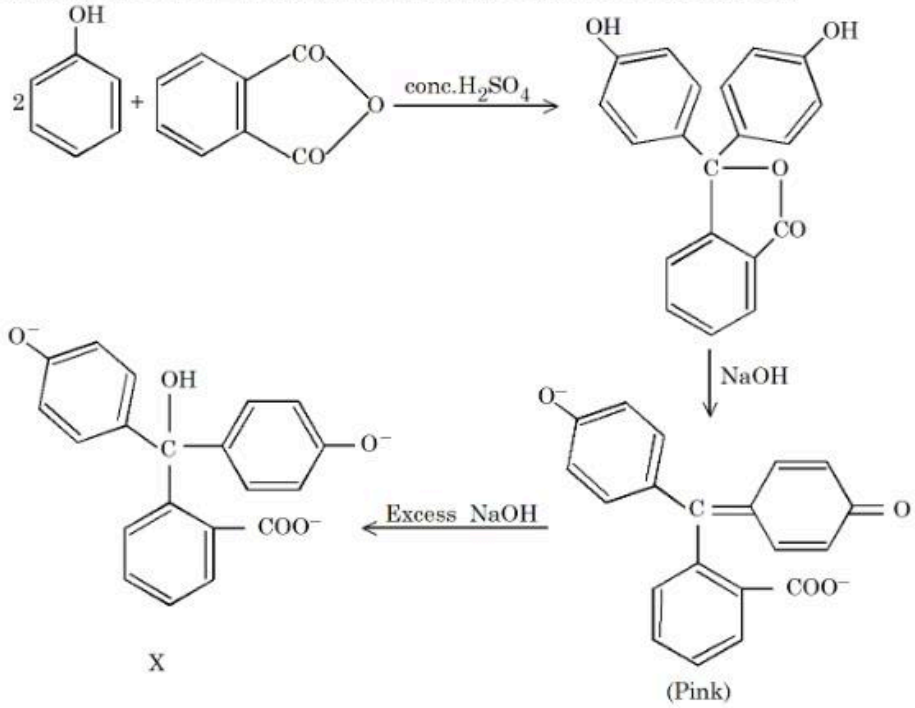
6952781009. A-IV, B-III, C-II, D-I

6952781010. A-III, B-IV, C-I, D-II

6952781011. A-II, B-IV, C-I, D-III

Question Number : 70 Question Id : 695278295 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the colour of compound 'X' in the sequence of the reaction.



Options :

6952781012. Violet

6952781013. Green

6952781014. Red

6952781015. Colourless

Chemistry Section B

Section Id :	69527824
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527824
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 695278296 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

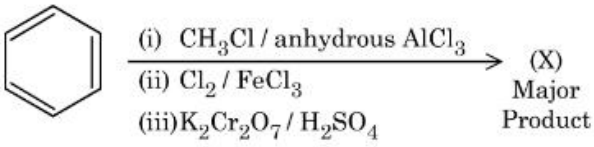
According to Lewis theory, the total number of σ bond-pairs and lone pair of electrons around the central atom of XeO_6^{4-} ion is _____.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 695278297 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following sequence of reactions to give the major product (X)



P g of the major product (X) formed is reacted with $NaHCO_3$ solution to liberate a gas which occupied 11.2 dm^3 at STP.
P = _____g.
(Given molar mass in g mol^{-1} H:1, C:12, O:16, Cl:35.5)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 695278298 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

2.0 g of a bromo hydrocarbon (X) was subjected to Carius analysis, gave 3.36 g of AgBr. The percentage of carbon in the compound (X) is 26.7%. Total number of carbon atoms in the empirical formula for compound (X) is _____.
(Given molar mass in g mol^{-1} H:1, C:12, Br : 80, Ag : 108)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 695278299 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The pH of a solution obtained by mixing 5 mL of 0.1 M NH_4OH solution with 250 mL of 0.1 M NH_4Cl solution is _____$\times 10^{-2}$.(Nearest integer)
Given: pK_b (NH_4OH) = 4.74

log 2 = 0.30
log 3 = 0.48
log 5 = 0.70

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 75 Question Id : 695278300 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A non-volatile, non-electrolyte solid solute when dissolved in 40 g of a solvent, the vapour pressure of the solvent decreased from 760 mm Hg to 750 mm Hg. If the same solution boils at 320 K, then the number of moles of the solvent present in the solution is _____. (Nearest integer)
[Given: boiling point of the pure solvent = 319.5 K,
K_b of the solvent = 0.3 K kg mol^{-1}]

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1